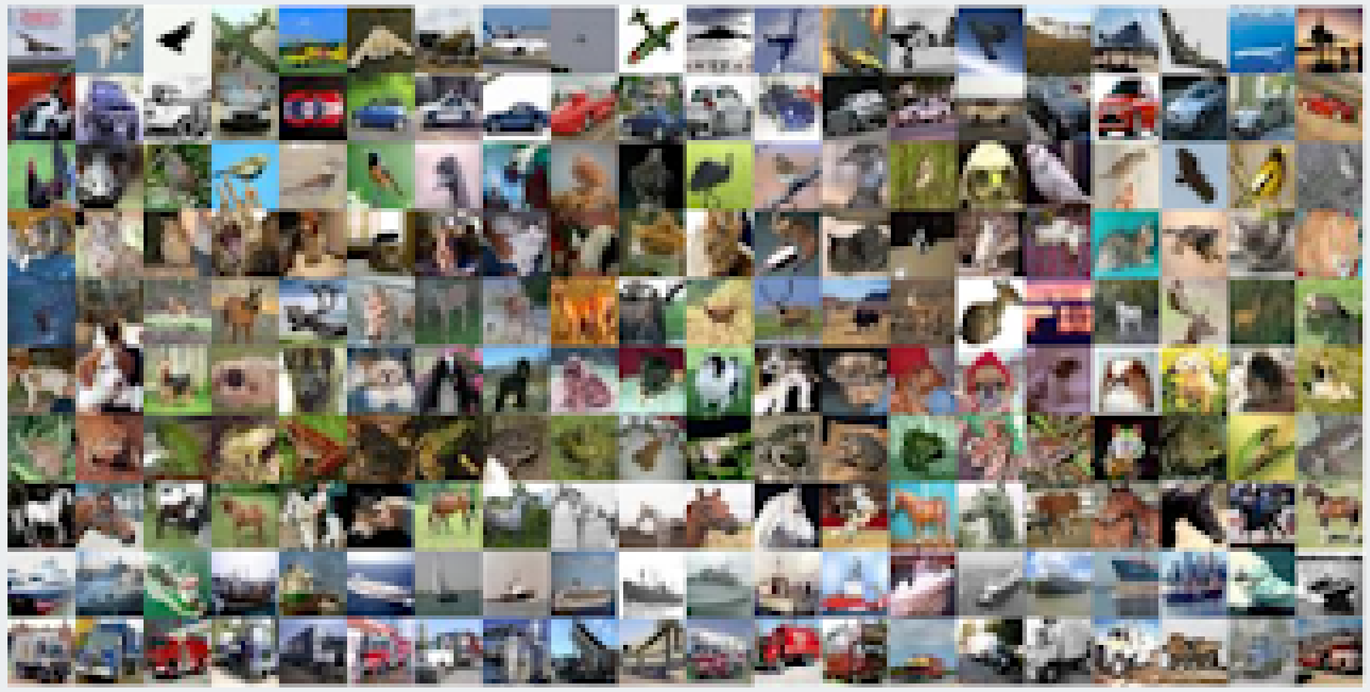
ImageNet Classification with Deep Convolutional Neural Networks

Conference: 2012
Dataset: ImageNet

Presenter: Jiwon Kang

Introduction

MNIST: 10 classes



ImageNet: 1000 classes

문제

- Large-scale image classification
- 1000 object classes
- millions of images

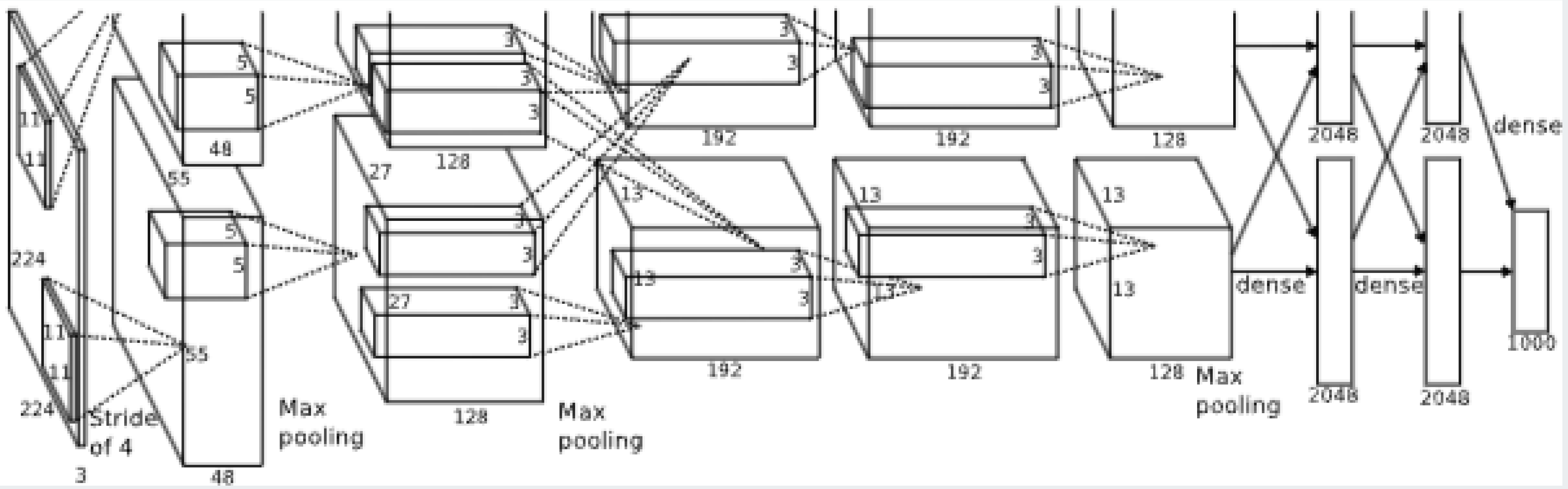
기존 방법

- Hard-crafted features
- shallow classifiers

한계

- comlex patterrns 학습 어려움

Background: CNN



CNN: Convolutional Neural Network

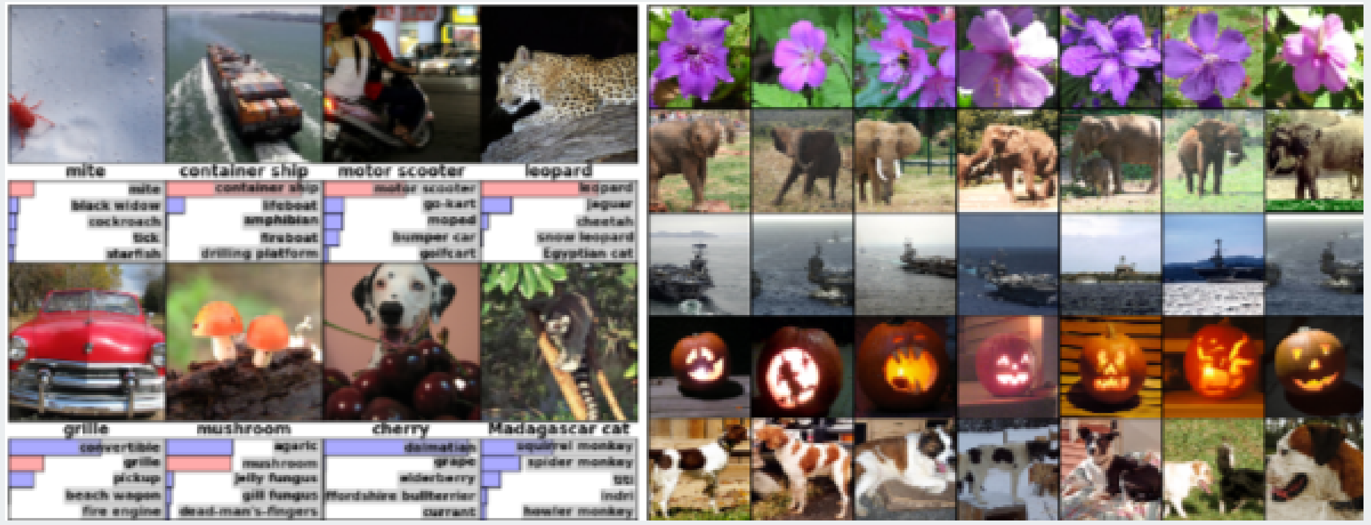
특징

- Local receptive field
- Weight sharing
- Pooling
- Hierarchical feautre learning

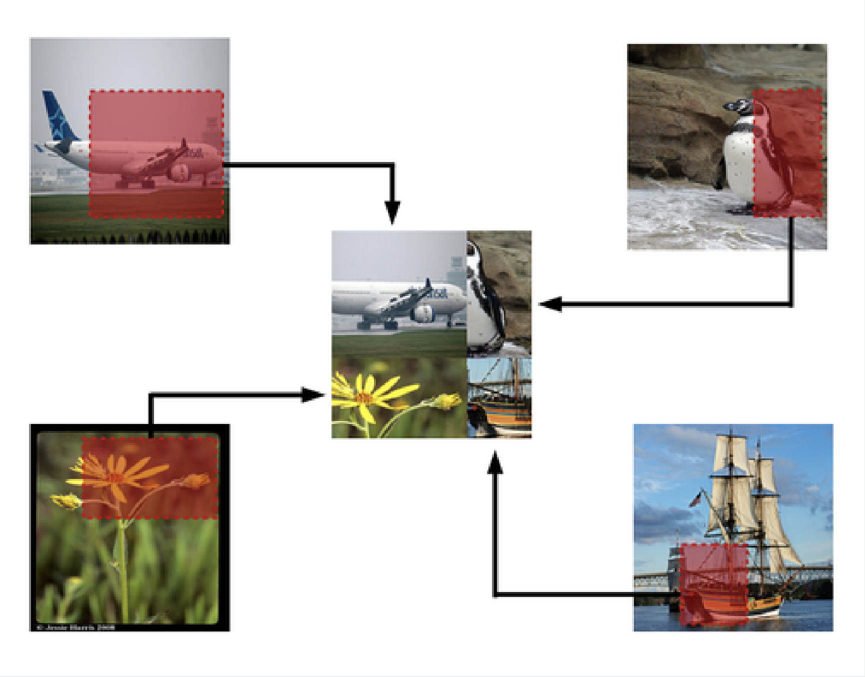
장점

- Spatial information 유지
- Feature 자동 학습

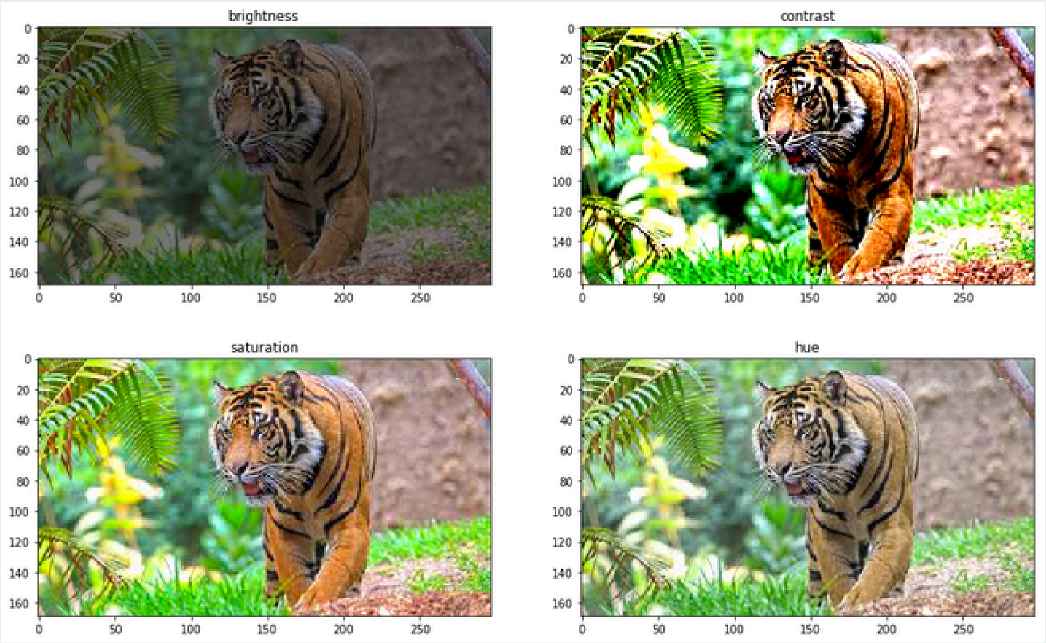
Dataset: ImageNet



ImageNet



Preprocessing: Random Crop



Preprocessing: Color Augmentation

ImageNet

- 1.2M training images
- 50k validation images
- 150k test images
- 1000 classes

- ILSVRC (ImageNet Large Scale Visual Recognition Challenge)

Preprocessing

- Images resized to 256×256
- Random 224×224 crops
- Mean subtraction
- Data augmentation (flip, color jitter)

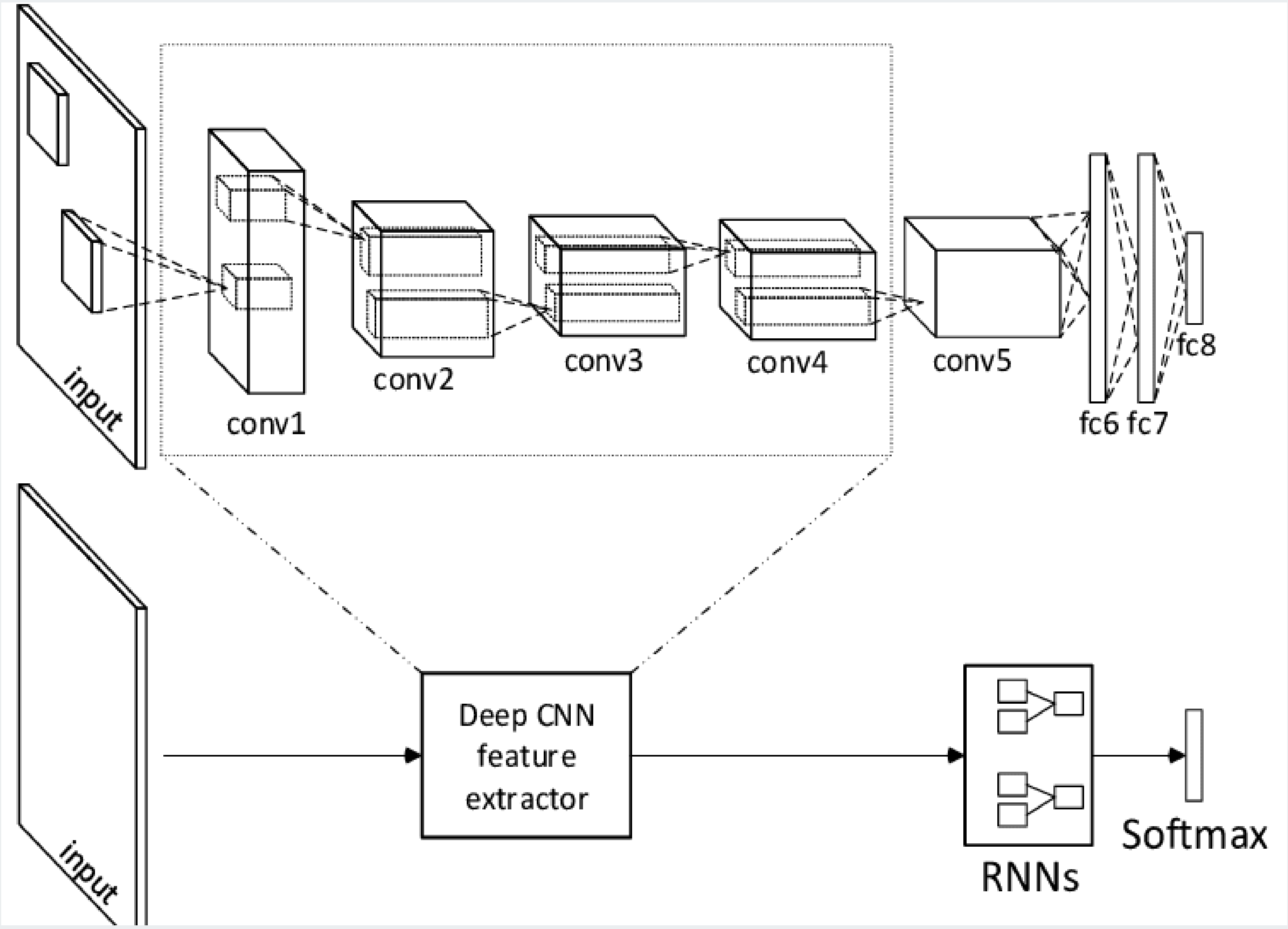
그래서 왜 Deep CNN이 필요했는가?

- Large dataset
- comlpex object variation
- shallow networks

왜 AlexNet 이전에는 Deep CNN이 잘 작동하지 않았는가?

- GPU 부족
- 데이터 부족
- activation 문제 (ReLU 이전)

AlexNet Architecture



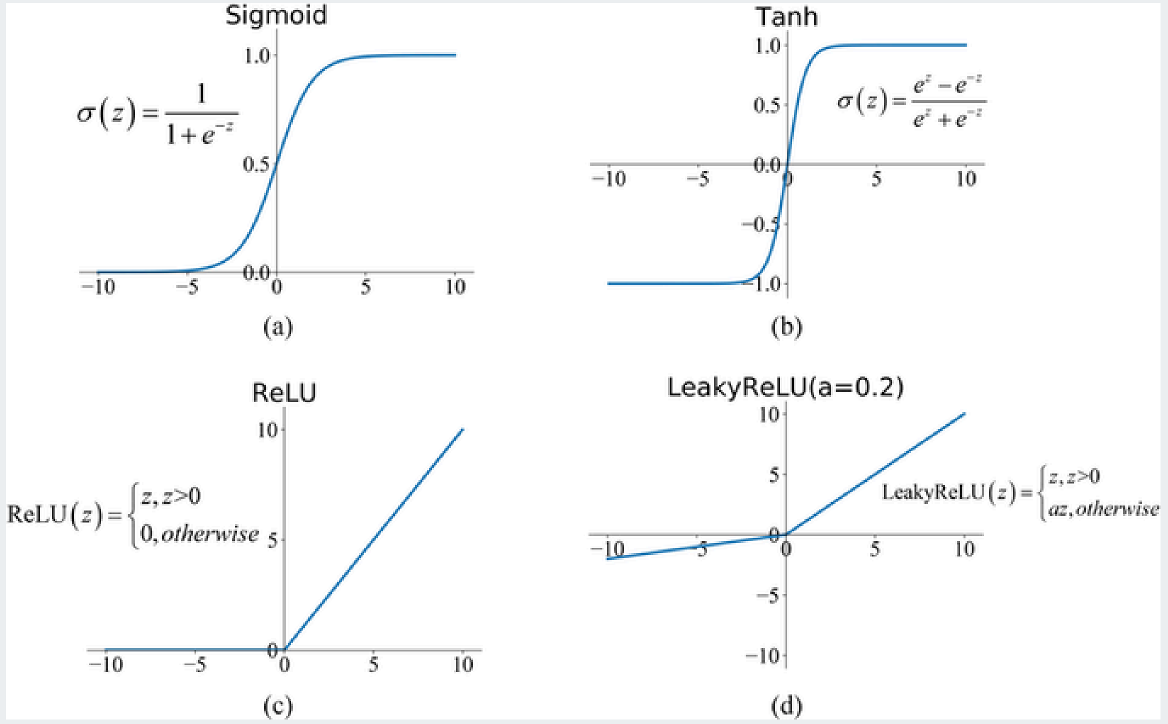
AlexNet

Network Flow

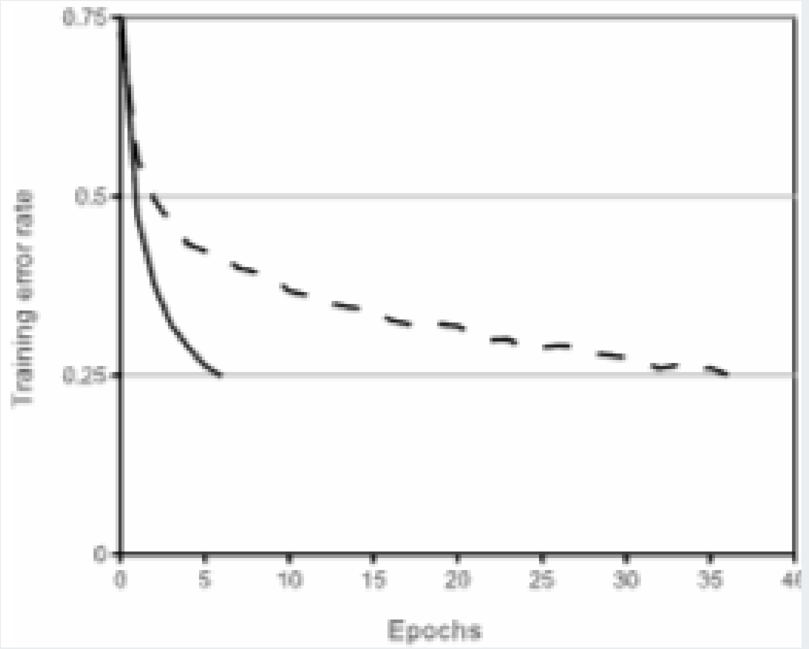
Input Image
→ Conv1
→ Conv2
→ Conv3
→ Conv4
→ Conv5
→ FC6
→ FC7
→ FC8
→ Softmax

- ~60M parameters
- 8 learnable layers
- 5 Convolution layers
- 3 Fully Connected layers

ReLU Activation



Activation functions



ReLU vs Tanh

기존 Activation

- Sigmoid
- tanh

→ vanishing gradient, slow training 문제

ReLU (Rectified Linear Unit)

$$f(x) = max(0, x)$$

- faster convergence
- simple computation
- sparse activation

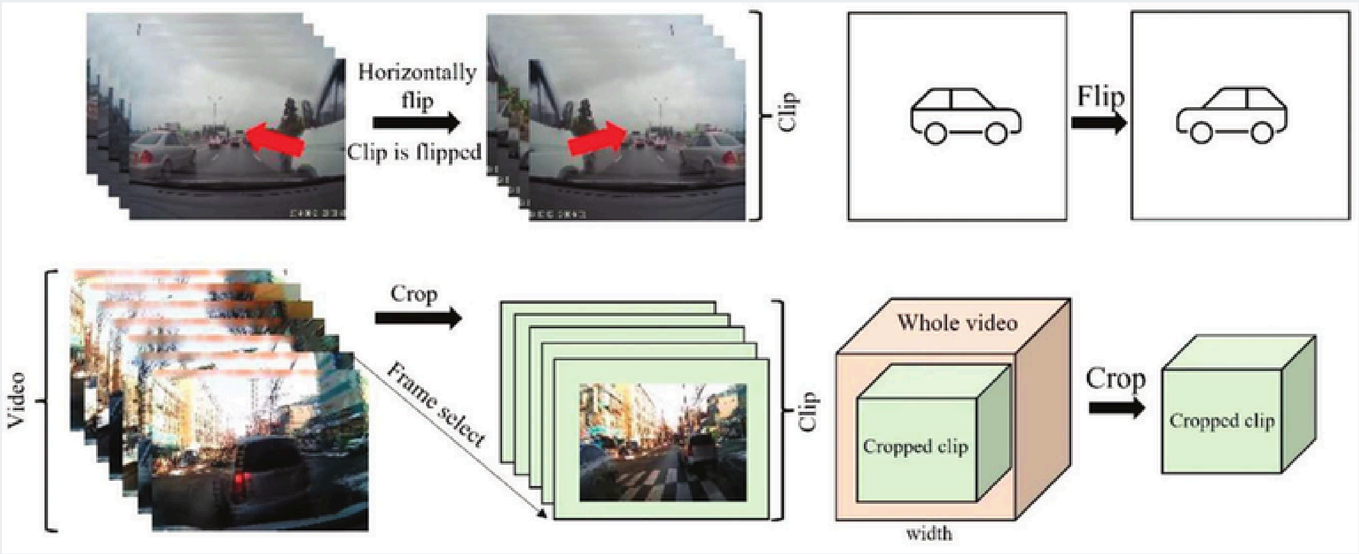
Reducing Overfitting in AlexNet

Overfitting Problem

- large number of parameters (~60M)
- limited labled data

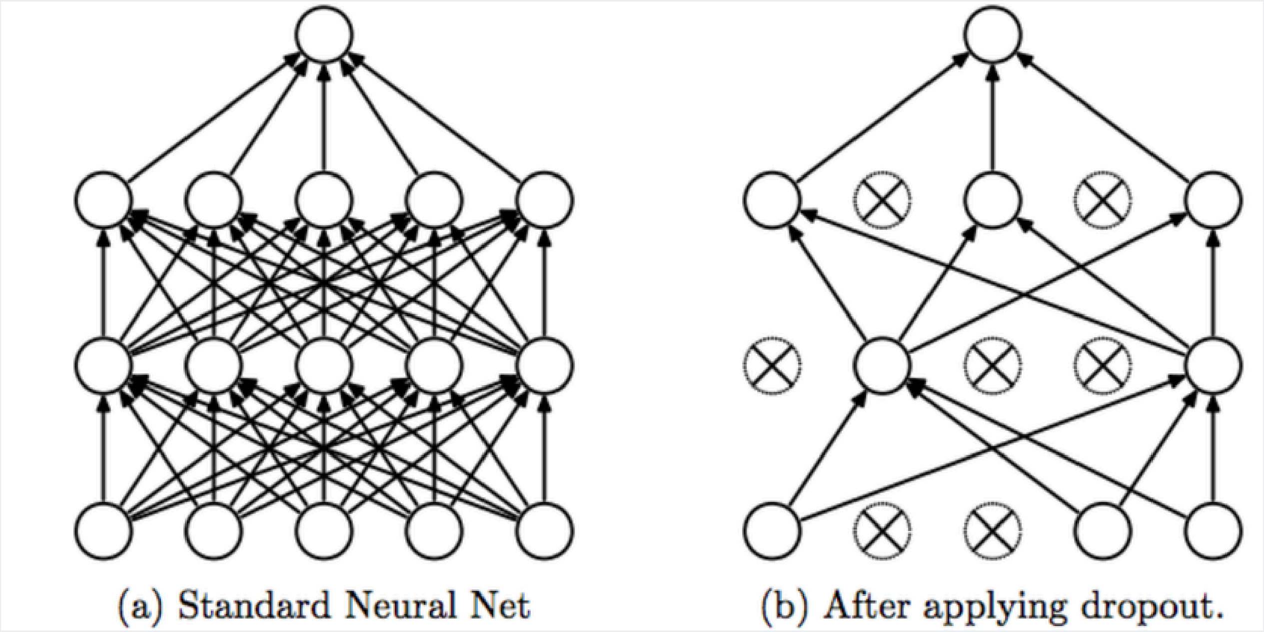
Data Augmentation

- **random crop**
- **horizontal flip**
- **color augmentation**



Dropout

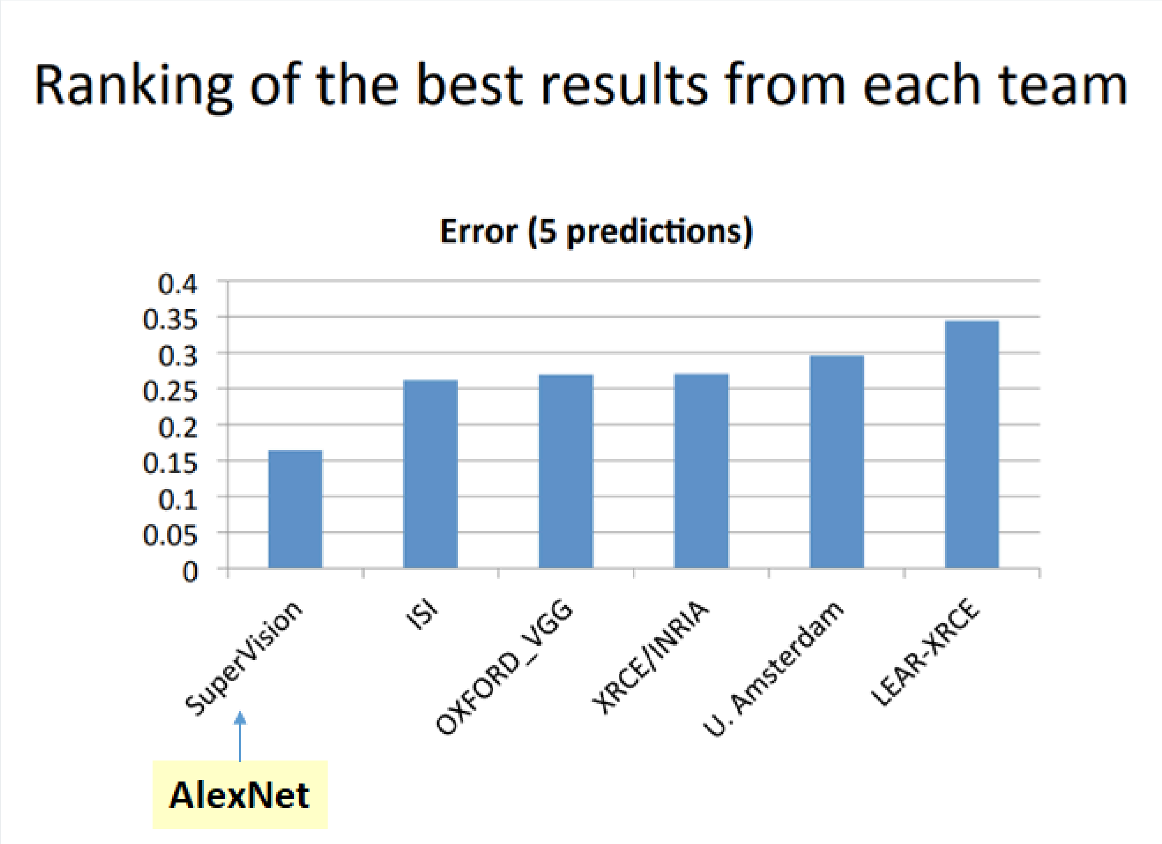
- **randomly deactivate neurons during training**
- **prevents co-adaptation**



Results & Significance

Experimental Results

Model	Top-1 (val)	Top-5 (val)	Top-5 (test)
SIFT + FVs [7]	—	—	26.2%
1 CNN	40.7%	18.2%	—
5 CNNs	38.1%	16.4%	**16.4%**
1 CNN*	39.0%	16.6%	—
7 CNNs*	36.7%	15.4%	**15.3%**



ImageNet ILSVRC 2012

Method	Error
Previous best	26.2%
AlexNet	**15.3%**

➡ 약 10% 이상 성능 향상

Significance

- First successful deep CNN on large-scale vision task
- Triggered deep learning revolution in computer vision
- Influenced later architectures

Examples

- VGGNet
- GoogLeNet
- ResNet

Conclusion

Key Takeaways

- • AlexNet: deep CNN for large-scale image classification
- • Training techniques: ReLU, Data augmentation, Dropout
- • Large performance improvement on ImageNet

Impact

- • Start of deep learning era in computer vision
- • Inspired many later CNN architectures

Thank You for Listening